

# **Stat 414 Notes**

# Table of contents

|                                                                 |           |
|-----------------------------------------------------------------|-----------|
| <b>Course Goals for STAT 414</b>                                | <b>3</b>  |
| <b>1 Introduction to Sampling</b>                               | <b>4</b>  |
| 1.1 Introduction . . . . .                                      | 4         |
| 1.2 Considerations in planning and executing a survey . . . . . | 8         |
| <b>2 An Introduction to Probability Samples</b>                 | <b>16</b> |
| 2.1 Probability Sampling Definitions and Notation . . . . .     | 17        |
| 2.2 Probability Samples: Inference Basics . . . . .             | 19        |

# Course Goals for STAT 414

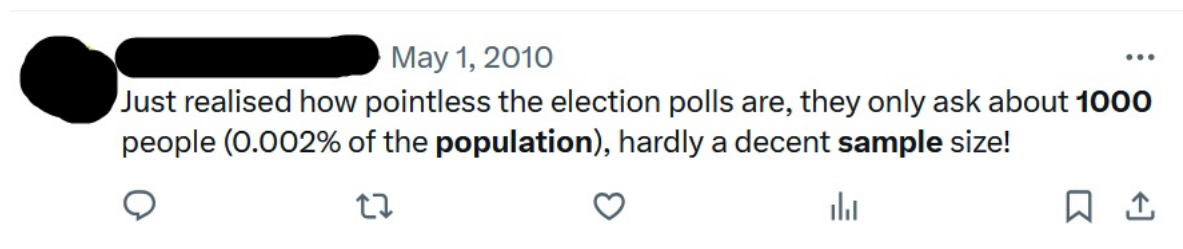
STAT 414 is an introduction to sampling. By the end of the course, you will:

- Learn the basic methodology of how classical survey samples are conducted based on simple random sampling.
- Recognize sampling and non-sampling errors.
- Be able to use extra information in sampling via ratio estimators and regression estimators.
- Be able to define and use stratified samples and know why they are often better than SRS.
- Be able to design one and two stage cluster samples and know their key properties.
- Recognize strengths and weaknesses of different sampling approaches.
- Use standard R packages to analyze survey data and estimate population means, proportions, totals, and more.

# 1 Introduction to Sampling

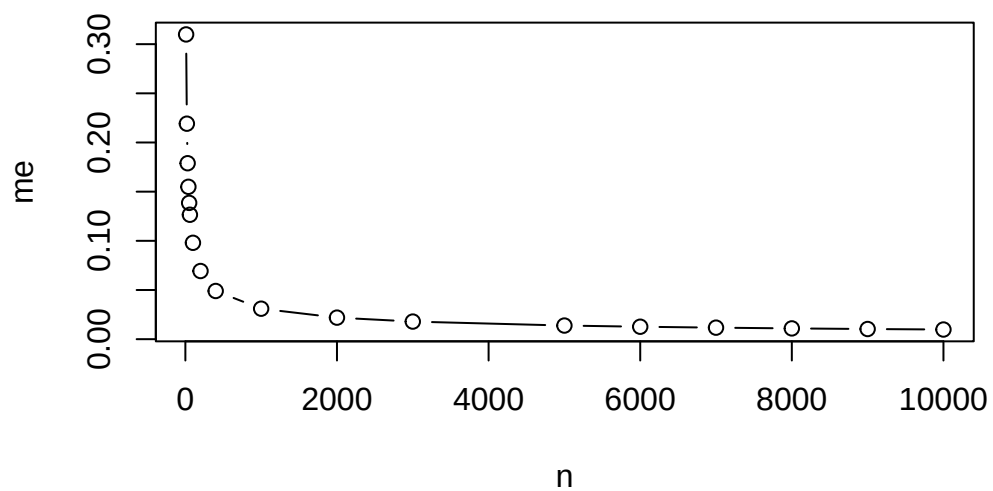
## 1.1 Introduction

If you spend any time at all on social media (especially in an election year), you're likely to come across a post like this:



What do you think? Do you agree?

```
n<-c(10,20,30,40,50,60,100,200,400,1000,2000,3000,5000,6000,7000,8000,9000,10000)
me<-1.96*sqrt(0.25/n)
plot(n,me,type="b")
```



So why even sample?

1. Cheaper

2. Faster

3. **More Accurate** (?!?!?)

  Aug 26, 2020 ...

Wednesday open question: What's one thing that's very hard for most people to believe, but which is actually true?

 414    101    241         

  ...

a surveyed sample of 1,000 individuals, if representative of the larger population, is sufficient to make accurate predictions about a population of 300 million people

3:18 PM · Aug 26, 2020

What's key here?

What we will spend this semester studying is how to ensure this, to make sure we can make accurate predictions. This will involve answering questions focused on:

- Survey design
- Preparation
- Collect and prepare data
- Data analysis

How is sampling different from previous courses, like STAT 212?

How is sampling different from previous courses, like STAT 262 or STAT 380?

How is sampling different from previous courses, like STAT 102 or STAT 318?

However, like previous courses, sample surveys do have an overarching process, some of which overlap with steps/processes we've seen before. We'll go through the principal considerations in a sample survey, along with some definitions and (possibly) new notation.

## 1.2 Considerations in planning and executing a survey

1. Objectives of the survey
2. **Defining the population** In sampling, the word **population** needs some clarification. Generally, it means the larger set from which the sample is chosen, and it's the larger group we wish to make inference/draw conclusions about. And a **sample** is a subset of the population. However, we must distinguish between
  - **Target population**
  
  - **Sampled population**

Ideally, these two overlap completely, but this does not always occur. Plus, we also need to consider the **sampling frame population**

I love the figure on page 5 of the book:

Within the **sampling frame** we have

- **Observation unit**
- **Sampling unit**

For example,

3. **Identifying potential sources of bias and ameliorating them:** In previous classes, like STAT 262 or STAT 380, we use the word **bias** to mean an estimator that systematically over- or under-estimates the true population characteristic. In sample surveys, we still use the same definition, but now biased estimators typically arise for two reasons:

- Population characteristic of interest (Estimation bias):
- Selection and/or measurement bias: Selection bias occurs when there is a mismatch between the sampled population and the target population. Measurement bias occurs when the survey observation tends to differ from the truth in a particular direction.

We'll talk about each of these in more detail.

## Selection bias

- Convenience samples
- Judgment samples
- Self-selected samples
- Undercoverage
- Overcoverage
- Nonresponse

## Measurement bias

Why can it occur?

- Unavoidable Reasons
- People!
- Survey mode: method used to distribute and gather responses

So we're concerned about response bias—what now? Careful question design is key. We could spend weeks on just questionnaire design, but will only do a very brief overview. There are other resources on campus that can help with this aspect of surveys (Bureau of Sociological Research).

We'll start with types of questions to avoid, and then move on to other considerations when writing questions.

**Common types of biased questions:**

- Leading questions

How to fix it:

- Assumptive questions

How to fix it:

- Double-barreled questions

How to fix it:

- Double negatives

How to fix it:

**Other considerations:**

- Pay careful attention to answer scale options

- Check formatting
- Avoid jargon, and be specific
- Open or closed questions?
- Question order matters

Probably most important, test questions ahead of time with the target population.

#### **4. Choosing the sampling design/data collection approach**

## 5. Choosing the analysis approach

## 6. Data analysis

7. **What's next?** The more information we have about a population, the easier it is to devise a sampling plan that will give accurate estimates. Any past survey can serve as a guide for future samples, as it will provide information about variability and cost.

**The rest of the semester**

## 2 An Introduction to Probability Samples

Before we can jump into sampling designs, we need to establish a few basic concepts and notation. The sampling designs we're going to discuss are **probability samples**.

💡 Definition: Probability sample

A **probability sample**

Throughout this semester we'll discuss three different ways to design a probability sample, as well as the properties of estimators arising from these designs. Probability samples allow us to avoid selection biases and help ensure the sample is representative of the population, as probability samples take selection choice away from the researcher, and put it in the hands of random selection. The three major designs we'll discuss are:

- Simple random sampling
- Stratified random sampling
- Cluster sampling

We'll also consider **systematic sampling**

To simplify things, we'll make the following assumptions:

Depending on how far we get, we may revisit relaxing some of these assumptions at the end of the semester.

In all cases, our goal is to learn about some characteristic of the target population

and we'll use our sample data to estimate that characteristic. We'll also estimate standard errors, and use those to construct confidence intervals for the unknown population characteristic. We'll also discuss determining the necessary sample size.

## 2.1 Probability Sampling Definitions and Notation

Throughout this semester, we'll assume a finite population, or **universe** of  $N$  units:

Any sample we select from  $\mathcal{U}$  will be a subset of  $\mathcal{U}$ , and we'll denote the specific sample chosen as  $\mathcal{S}$ .  $\mathcal{S}$  contains  $n$  elements from  $\mathcal{U}$ .

**Example:** Suppose  $\mathcal{U} = \{1, 2, 3, 4, 5\}$  and we consider samples of size  $n = 3$ . There are 10 possible samples that could be selected:

Remember that in **probability sampling**

Because each possible sample  $\mathcal{S}$  has a known probability of being selected, each unit in the universe  $\mathcal{U}$  also has a known probability of being in the selected sample.

Let  $y_i$  be

We consider  $y_i$  to be a *fixed quantity*.

## 2.2 Probability Samples: Inference Basics

We want to use the data from a sample to estimate some characteristic of the population. Common population characteristics of interest are

**Example:** Earlier we considered the universe  $\mathcal{U} = \{1, 2, 3, 4, 5\}$ . Suppose the values of  $y_i$  are

|       |   |   |   |   |   |
|-------|---|---|---|---|---|
| $i$   | 1 | 2 | 3 | 4 | 5 |
| $y_i$ | 2 | 4 | 4 | 6 | 6 |

We also considered taking samples of size  $n = 3$  from  $\mathcal{U}$ . There are 10 possible samples, and all are equally likely.

Just like in other classes like STAT 102, 380, 212 our inference methods are going to rely on the **sampling distribution** of a statistic.

 Definition: Sampling Distribution

The **sampling distribution**

**Example, cont.:**

From the sampling distribution, we can find properties of the estimator we're using.

- **Expected value:**

- Estimation bias:

- Variance:

- Mean squared error: